Robustness and Representation Stability in Environmental Sound Classification: MFCCs versus Pretrained YAMNet Embeddings

Final Project Report

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GitHub Project Site

https://ysha0293.github.io/elec5305-project-530087661/

GitHub Repository

https://github.com/ysha0293/elec5305-project-530087661

Project Video

https://drive.google.com/file/d/1oTqImjyqKpDkdofGKj4SkjYRO1NiDER3/view?usp=drive_link

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1 Introduction

1.1 Problem and Motivation

The Environmental Sound Classification (ESC) aims to recognize the different acoustic events in the real environment, like Sirens, gunshots, drilling sounds and street music, etc. UrbanSound8K is one of the most common datasets in this field, including: 10 types of urban environmental sounds and providing the official 10 fold split; and it provides a unified evaluation basis for different classification methods [1].

The traditional methods usually use handcrafted acoustic features, like Mel-Frequency Cepstral Coefficients (MFCCs); but in recent years, the pretrained deep audio model is applied to extract the higher-level audio representations widely. YAMNet is pretrained based on the large-scale AudioSet data, which can be used as a frozen feature extractor to generate deep audio embeddings[2].

However, the higher benchmark classification performance doesn't mean that the system can be stable in the real recording environment. The background noise, reverb, and changes in the frequency response of the microphone or transmission channel may change the time-spectral structure of audio signal and further affect the classification performance. The existing study shows that the acoustic disturbance not only can cause the decrease of the down-stream classification performance, but also make the audio representation itself to change; therefore, the stability of representation can be an important perspective for studying the robustness of classification [3]. Hence, the main point of this project does not further pursue a higher UrbanSound8K benchmark accuracy, but to compare the stability of the traditional MFCC features and Pre-trained YAMNet representations when the recording condition is changed.

1.2 Main Research Question

Research Question: how robust are the traditional MFCC features and pre-trained YAMNet audio embeddings under acoustic and recording-channel perturbations in the environmental sound classification?

In order to answer this question, this project initially used the UrbanSound8K official ten fold structure to build the classification benchmark in the clean condition, then only the held-out test data are added some different strength background noise, reverberation and channel bandwidth distortion. By comparing the Macro-F1, classification recall and confusion patterns, analyze the performance degradation of two representations in the different disturbance conditions.

In addition, this project also conducts an auxiliary analysis:

Secondary analysis: Is the representation change between clean audio and disturbed audio related to the decline in classification performance?

For that, calculate the representation distance between the MFCC representations and YAMNet embeddings before and after adding the disturbance respectively, and analyze the correlation between the average representation shift and the decline of the Macro-F1.

1.3 Project Scope and Contributions

This project compares two main systems: MFCC statistical features combined with support vector machines, and the frozen pre-trained YAMNet audio embeddings combined with SVM. Both systems use the UrbanSound8K official ten fold evaluations, and the classifier only uses the clean training data to train. Background noise, reverberation, and channel distortion are added only to the testing data. At the same time the same disturbance audio is used for MFCC and YAMNet feature extraction, which can ensure the consistency in comparison.

In addition to the whole classification performance, this project also analyzes the sensitivity of different sound categories to various disturbances and combines the confusion matrix and typical

signal representations to explain some partial classification faults. Finally, by analyzing the correlation between the representation distance and the decline of the classification performance, to further study the relationship between representation stability and robustness.

1.4 Report Organisation

The second section summarizes some researches on the sound and environment related to above mentioned audio representation techniques such as environment sound classification, MFCC, pre-trained audio representation and robustness analysis. The third section introduces the dataset, experimental protocol, classifier, disturbance design and represents the distance analysis method. The forth section shows the experimental results under the clean condition and different disturbance conditions. The main finds, differences at the category level, representation stability and research limitations are discussed in the fifth section. The sixth section is the full text and presents future research directions.

2 Literature Review

2.1 Environmental Sound Classification and UrbanSound8K

Environmental Sound Classification (ESC) aims to classify environmental sounds of different types from the audio signals. One of the most widely used datasets in this area is UrbanSound8k, which contains: 8732 audio segments (no longer than 4s) and types of 10 typical urban sounds [1]. This dataset also provides official 10 folds split, which allows to be used for comparison of different classification methods with a unified experimental protocol.

Existing researches have explored the traditional machine learning and deep learning methods on the UrbanSound8k. For instance, Piczak has trained the convolution network to learn the time-spectral representation of environmental sounds and demonstrated that deep learning models can attain the competitive classification performance [4]. The goal of this project, however, is not to further improve the accuracy of the clean benchmark, but rather to examine if the feature representations can still be used to classify the data when the recording condition is altered.

2.2 Conventional MFCC Representations

Mel-Frequency Cepstral Coefficients (MFCCs) is a typical artificially designed acoustic representation [5]. The initial processing steps of its basic processing are: short-time frame division, window function processing, spectral calculation, Mel filter, Log Compress, Discrete Cosine Transform. In this project, this process has the clear meaning of digital signal processing (DSP), and MFCC is appropriate for traditional and interpretable baseline.

The great advantage of MFCC is that it is easily understandable, simple to compute, and it can efficiently describe the short-time spectral envelope. The background noise, reverberation and frequency response, however, may impact the MFCC representation due to the direct dependency of the MFCC on the spectral energy distribution of the input signal. MFCC is therefore used as the appropriate baseline to work on the stability of the handcrafted audio representation in various recording conditions.

2.3 Pretrained Audio Representations and YAMNet

In recent years, large-scale pre-trained audio models have gradually been used for the down-stream audio classification task. AudioSet provides the large scale real world audio event data, which provides baseline for learning common audio representation [2]. YAMNet is an audio classification network pre-trained based on AudioSet and its inner embeddings can be used for feature representation of other audio tasks.

This project treats YAMNet as the frozen feature extractor, rather than re-training the network in the UrbanSound8K. YAMNet embedding and MFCC features all use the simple SVM as the down-stream classifier; therefore, the focus of the comparison can be concentrated on the representation itself, rather than the difference in the complexity of classification models.

2.4 Robustness to Acoustic and Channel Perturbations

The high performance of the standard dataset doesn't mean that the system can adapt the real recording condition changes. The background noise, reverb and the frequency response of the microphone or transmission channel all may change the time spectral structure of input audio, which can affect the feature representations and final classification results.

Srivastava et al studied the change of the environmental sound representations under different channel-like perturbations, and used YAMNet and OpenL3 to analyze the distance between the original and disturbance embedding as well as the downstream classification performance [3]. This study simulated the recording channel changes by disturbances such as high pass, low pass, gain and reverberation, and pointed that the change of audio embedding and downstream performance degradation can be combined to evaluate the representation robustness.

This research also shows that only relying on the classification performance cannot fully describe the changes that occur in representation under perturbation; Therefore, it can further directly measure the distance between the original and disturbance representation. Its results show that the different perturbations have different effects on the representation space and some distance-change trends have an obvious relationship with the decline of down-stream performance.

2.5 Gap and Method Selection

Existing research has proved that it is worth analyzing the robustness of pretrained audio embeddings solely, but it is relatively rare to compare the traditional handcrafted features and modern pretrained representations directly from the perspective of the ELEC5305 signal processing. Therefore, this project chooses the MFCC and YAMNet as two representation methods with clear differences, and conducts a comparison under the unified SVM classifier and the official ten-fold UrbanSound8K protocol.

Unlike only comparing clean accuracy, this project focuses on evaluating performance changes under the background noise, reverb and channel bandwidth distortion. At the same time, by class-level recall, confusion matrix and clean-to-perturbed representation distance, further analyze the sensitivity of different categories and the relationship between the representation changes and the decline of classification performance. This design also can connect the traditional DSP features, pre-trained representations and robust analysis at the same time, and directly serves the main research problem of this project.

3 Methods

Figure 1 summarises the overall experimental workflow used in this project.

3.1 Dataset and Official Evaluation Protocol

This project uses the 10-fold partition officially provided by the UrbanSound8K to conduct all experiments. For the f-th fold, fold f is used as the held-out test set, and other nine folds are used for training dataset. This process repeats ten times, making each fold exactly serve as a test set once.

For each fold, train dataset and test dataset are all determined by the official fold label and do not do other partitions. Classifier training, feature standardization parameter estimation and other relevant training operations only use current nine training folds. The corresponding test fold only is

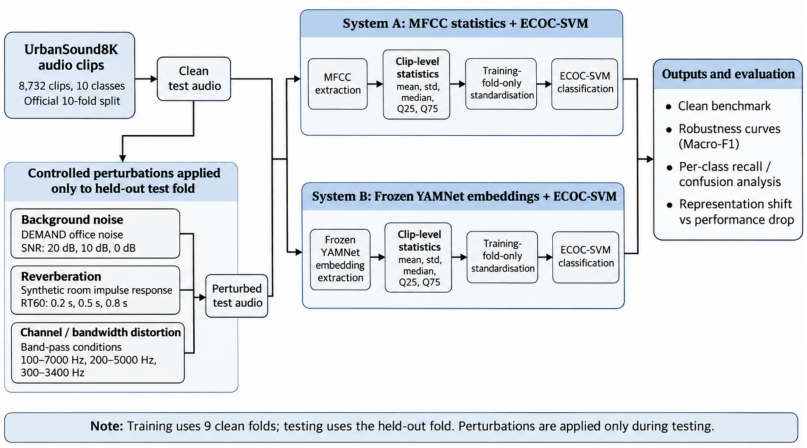


Figure 1: Overall experimental workflow for comparing MFCC and frozen YAMNet representations under controlled acoustic and channel perturbations.

used for final prediction and performance computation.

The final classification performance is summarised across the ten folds, and report the Accuracy and the average value and standard deviation of Macro-F1.

3.2 System A: MFCC Statistics with SVM

For MFCC system, each audio clip is first subjected to uniform preprocessing and transforms to the single channel audio that is suitable for feature extraction. After that, extracting 13 dimensions MFCC and corresponding 13 dimensions delta features. Therefore, each time frame includes 26 feature values.

In order to make Frame-level features of different lengths transform to the fixed-length representation, this project respectively calculates five statistics for each feature dimension:

$$\text{mean}, \quad \text{standard deviation}, \quad \text{median}, \quad Q_{25}, \quad Q_{75}$$

Therefore, each audio clip finally obtains 26 * 5 = 130 dimension MFCC Statistical eigenvector.

The classification step applies the multi-class classification framework of MATLAB, and uses the linear SVM as the binary classification learner. For each official fold, only using 130-dimension feature classifier of the corresponding nine training folds.

Before training, the standardized mean and standard deviation only are calculated by the training dataset. The test features are transformed using the same training statistics:

$$x' = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

This process ensures the test data does not participate in the estimation of standardized parameters

3.3 System B: Frozen YAMNet Embeddings with SVM

For YAMNet system, each audio clip is preprocessed according to the input requirements of the pre-trained YAMNet, and is extracted frame-level embeddings. YAMNet remains frozen throughout the experiment without retraining or fine-tuning.

After that, calculate the mean, standard deviation, median, (Q_25) and (Q_75) for every embedding dimension; and concatenate them to form a clip-level representation of a fixed length.

The obtained YAMNet representation uses the same SVM classification process as the MFCC system. For every official fold, classifier only uses the remaining nine clean training folds to conduct

training, and normalisation parameters also uses the training folds to calculate and is applied to the corresponding held-out test fold.

Therefore, two systems keep the consistency in the aspect of data partition, standardization and classifiers; The main difference of these two systems is from the audio representation itself.

3.4 Clean Benchmark

Before adding any disturbance, initially using the original UrbanSound8K audio to build clean benchmark. MFCC-SVM and YAMNet-SVM all follow the same official ten-folds protocol to evaluate and record the Accuracy, Macro-F1, per-class recall and confusion matrix of every folds. Finally using the average value and standard deviation of ten folds to summarize the overall performance, and using it as the following reference benchmark of robustness experiments

3.5 Controlled Perturbations

In order to simulate the real recording condition changes, this project only adds the disturbance to each fold held-out test audio and the training data always keeps original clean audio. Three kinds of disturbances respectively are background noise, reverberation and channel/frequency-response distortion. Three severity levels are set for each disturbance and same perturbed waveform is used for the MFCC and YAMNet feature extraction at the same time, ensuring the comparison conditions between the two systems are consistent.

3.5.1 Background Noise

The background noise is from the Diverse Environments Multichannel Acoustic Noise Database (DEMAND) that is an environmental noise database and independent of UrbanSound8K [6]. The experiment uses the *OOFFICE* office environment 16 kHz recording, uses the same fixed channel (ch01) as the noise source for all noise experiments.

Using the external independent noise source can prevent the classified samples in UrbanSound8K from being re-used as interference sources, and making background-noise perturbation closer to the real recording environment. The OOFFICE noise was added to each held-out test clip at target signal-to-noise ratios (SNRs) of

$$20 \text{ dB}, \quad 10 \text{ dB}, \quad 0 \text{ dB}.$$

The noise amplitude can be scaled according to the original signal power and the target SNR, so that different severity levels have a unified and controllable definition. 20 dB means the light noise, 10 dB means the moderate disturbance, and 0 dB represents a strong condition in which the signal and noise powers are approximately equal.

The noisy waveform generated for each test sample is used for MFCC and YAMNet feature extraction at the same time. DEMAND data source, specific channel and license information that are used in the experiment are recorded in the reproducibility metadata of the project.

3.5.2 Reverberation

The reverberation is implemented by convolving the held-out test audio and a synthetic room impulse response. The experiment sets three degrees of reverberation severity, and its approximate reverberation times are respectively:

$$RT_{60} = 0.2 \text{ s}, \quad 0.5 \text{ s}, \quad 0.8 \text{ s}.$$

A large RT_60 indicates a longer decay time of sound energy and therefore produces more temporal smearing. All room impulse responses use the fixed experiment setting and same reverberant

waveforms input into both MFCC and YAMNet systems simultaneously.

3.5.3 Channel / Frequency-Response Distortion

In order to simulate the frequency response changes of the microphone changes or the limited transmission bandwidth, this project produces the channel distortion by limiting the effective frequency range of the held-out test audio. Three severity degrees are set as:

100–7000 Hz,

200–5000 Hz,

300–3400 Hz.

With the decrease of the bandwidth, more low-frequency and high-frequency information are removed, which can change the frequency spectral structure of inputs. The same perturbed waveform generated under each condition is used in the MFCC and YAMNet feature extraction simultaneously.

3.6 Representation-Shift Analysis

To measure the impact of disturbances on feature representation, this project compares the representation distance in a same audio under the clean and perturbed conditions. For MFCC summary vectors, using the Euclidean distance; For YAMNet embeddings, using the cosine distance.

After that, for each perturbation condition, calculate the average representation shift and compare it with its corresponding Macro-F1 drop. Pearson correlation and Spearman correlation are used to evaluate the correlated degrees between the representation change and classification-performance degradation.

All models and normalizing parameters all only use the clean training folds to estimate, and all disturbances only are added to held-out test fold. At the same time, two systems all use exactly the same perturbed waveform under every experiment condition, ensuring the consistency of comparison and avoiding the data leakage.

4 Results

4.1 Clean Benchmark

First, the two systems were evaluated under the clean UrbanSound8K condition, to build the reference baseline of the following robustness experiments. Based on the official ten-folds results, the average Accuracy of MFCC-SVM is 0.6343, Macro-F1 is 0.6423; The average Accuracy of YAMNet-SVM is 0.7837, Macro-F1 is 0.7918. The corresponding ten-folds standard deviation are 0.0482, 0.0465, 0.0319 and 0.0370 respectively. Table 1 summarizes the whole classification performance of these two systems under the clean condition. It indicates that YAMNet-SVM is significantly superior to MFCC-SVM in both Accuracy and Macro-F1, and the fluctuation across folds is relatively smaller.

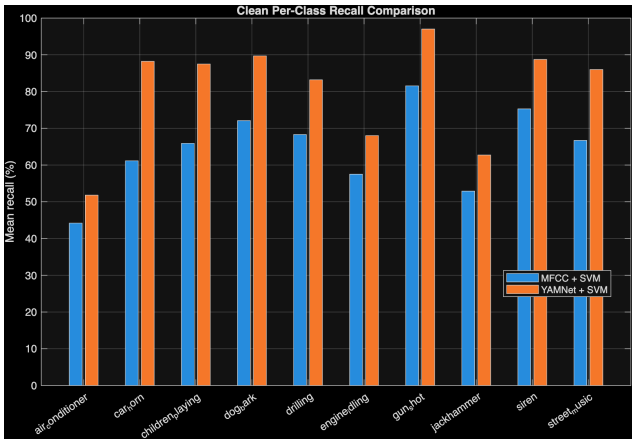
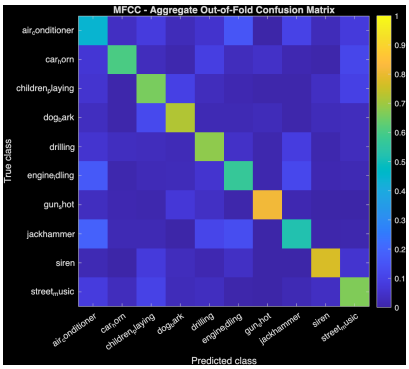


Figure 2: Per-class recall of the MFCC-SVM and YAMNet-SVM systems under clean UrbanSound8K conditions.

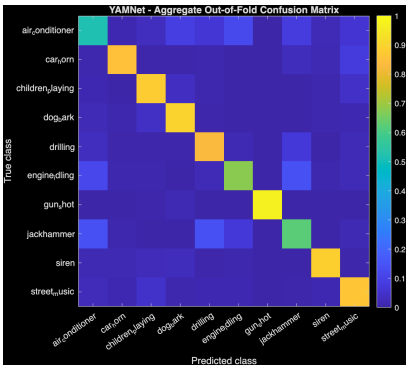
Table 1: Clean-condition classification performance across the official ten UrbanSound8K folds.

Method	Accuracy	Macro-F1
MFCC-SVM	0.6343 ± 0.0482	0.6423 ± 0.0465
YAMNet-SVM	0.7837 ± 0.0319	0.7918 ± 0.0370

Figure 2 compares the average recall of the two systems in ten sound categories. The recall of YAMNet-SVM in all categories is higher than that of MFCC-SVM, and the differences in categories such as *car horn*, *children playing*, *dog bark*, *drilling*, *siren* and *street music* are quite obvious. *Gun shot* has higher recall in these two systems but the recall of the *air conditioner* and *jackhammer* are relatively lower, which indicates that there are still obvious differences in classification difficulty among different categories.



(a) MFCC-SVM



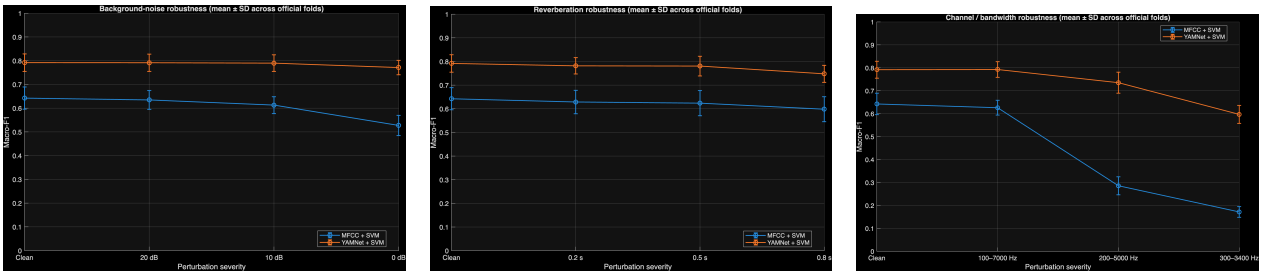
(b) YAMNet-SVM

Figure 3: Aggregate clean-condition confusion matrices across the official ten UrbanSound8K folds.

Figure 3 shows the normalized confusion matrices summarized by the official ten-folds testing results of two systems. The whole results of the YAMNet-SVM are generally more concentrated on the main diagonal, which is consistent with the higher per-class recall and Macro-F1. In comparison, the MFCC-SVM performs more obvious non-diagonal prediction in multiple categories, which indicates that the classification errors of its clean condition are more scattered.

In total, the clean benchmark shows that frozen YAMNet embeddings can provide stronger baseline classification performance than the MFCC statistics in the same down-stream SVM classification framework. This result can be the clean reference for the following experiments (noise, reverberation and channel distortion), which can be used for the performance degradation under different disturbance conditions.

4.2 Robustness under Changing Acoustic Conditions



(a) Background noise (b) Reverberation (c) Channel distortion

Figure 4: Macro-F1 as perturbation severity increases. These curves directly address the main research question by comparing the robustness of MFCC and YAMNet representations.

Figure 4 compares the Macro-F1 of the MFCC-SVM and YAMNet-SVM in the different disturbance intensities. The results show that the robustness difference between these two kinds of representations is closely related to the type of the specific disturbance.

For the background noise, the total change of the YAMNet-SVM is small under the different SNR conditions, its Macro-F1 decreases from 0.7918 under the clean condition to 0.7711 at 0 dB. However, the Macro-F1 of the MFCC-SVM decreases from 0.6423 to 0.5272. the corresponding performance degradations of the two systems are respectively about 0.0207 and 0.115, which indicates YAMNet is more stable against additional noise.

In the condition of reverberation, the Macro-F1 of both systems gradually decreases as the reverberation increases. Under the strongest conditions, the Macro-F1 of MFCC-SVM is 0.5982 and the Macro-F1 of YAMNet-SVM is 0.7479. Compared with the clean baseline, both of them decreased about 0.044. Therefore, although YAMNet-SVM still keeps higher overall classification performance, the robustness drops of the two systems are very close under the condition of reverberation.

The largest difference occurs under the condition of channel distortion. As the effective bandwidth decreases, the performance of the MFCC-SVM decreases obviously, reaches only 0.1716 under the strongest conditions; but YAMNet-SVM still keeps 0.5969. The corresponding Macro-F1 drops separately are 0.4707 and 0.1950, which demonstrates that the severe frequency response distortion has a more obvious impact on MFCC representation.

Table 2 summarizes the results of the three types of disturbances under the strongest conditions. In total, YAMNet shows stronger robustness in the background noise and severe channel distortion, but the performance drops of the two systems are relatively close under the condition of reverberation.

Table 2: Performance under the strongest condition of each perturbation family.

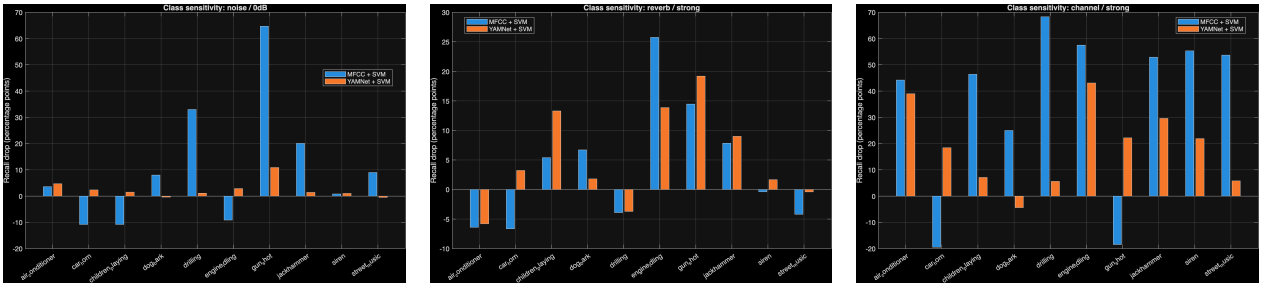
Perturbation	Condition	MFCC F1	Drop	YAMNet F1	Drop
Noise	0 dB	0.5272	0.1151	0.7711	0.0207
Reverb	0.8 s	0.5982	0.0441	0.7479	0.0440
Channel	300–3400 Hz	0.1716	0.4707	0.5969	0.1950

4.3 Class-Level Sensitivity

Figure 5 shows the recall drop of each sound category under three types of strong disturbance conditions. There are significant differences in the sensitivity of different categories to disturbances, and MFCC-SVM and YAMNet-SVM do not show their largest performance drop in the same category.

Under the condition of the strong noise, the MFCC-SVM is most sensitive to *gun shot*, the recall drop is close to 65 percentage points, and the performance of *drilling* and *jackhammer* occur obvious

decline. However, the recall drops of YAMNet-SVM are relatively small in each category, and the *gun shot* is still the most sensitive category.



(a) Strong noise (b) Strong reverberation (c) Strong channel distortion

Figure 5: Per-class recall degradation under the strongest level of each perturbation family.

In the condition of strong reverberation, the category sensitivities of the two systems are more dispersed. The performance drop of MFCC-SVM is most obvious in the *engine idling*, but the recall drop of the YAMNet-SVM is relatively large in the *gun shot* and *engine idling*. It indicates that the effect of reverberation to these two kinds of representations is not completely concentrated on the same category.

Strong channel distortion produces the most obvious category difference. MFCC-SVM shows significant declines in multiple categories such as *drilling*, *engine idling*, *siren*, and *street music*, and the drop of *drilling* is the largest. The recall degradation of YAMNet-SVM is generally small, but the result of the *engine idling* and *air conditioner* are still affected obviously.

In total, the category sensitivity has obvious perturbation-dependent features. Two systems are affected in the same category sometimes, but the main failure classes are not completely consistent, which indicates that MFCC statistics and YAMNet embeddings retain different information for different acoustic variations.

4.4 Representative Confusion Analysis

In order to analyze the specific fault mode under the severe channel distortion, this project chooses the strongest channel-distortion condition as the representative case. In the previous robustness analysis, two systems have the most obvious performance difference under this condition, so it is suitable to observe the category confusion and the change of the underlying signal representations.

Figure 6 compares the confusion matrices of the MFCC-SVM and YAMNet-SVM under this condition. The MFCC-SVM confusion matrix shows much stronger off-diagonal confusions, especially multiple categories are misclassified as *car horn* or *gun shot*, which indicates that the spectral information originally used to distinguish categories has been significantly weakened after the severe limitation of the bandwidth. However, the confusion matrices of the YAMNet-SVM still keeps relatively clear main diagonal structure, though there is still some confusion among classes such as *air conditioner*, *engine idling*, and *jackhammer*, the overall category separability is significantly better.

Figure 7 shows the waveform, spectrum, spectrogram and MFCC of two representative misclassified samples under the clean and perturbed condition. It shows that the strong channel distortion obviously weakens the high frequency area and changes the original spectral energy distribution and MFCC pattern. For the MFCC examples, *engine idling* is misclassified as *gun shot* after disturbance. For YAMNet example, *air conditioner* is misclassified as *dog bark*. It indicates that YAMNet is more stable, and the severe bandwidth limitation may still make the representation of some individual samples undergo sufficient changes, which causes wrong prediction.



(a) MFCC-SVM (b) YAMNet-SVM

Figure 6: Confusion matrices under the strongest channel-distortion condition.



(a) MFCC representation (b) YAMNet representation

Figure 7: Representative signal-level changes under strong channel distortion.

In total, this representative analysis shows that severe channel distortion may affect the underlying acoustic representation and final classification boundary. The category structure of MFCC-SVM has been more significantly disrupted, and the YAMNet-SVM still can keep more distinguishable information under the same disturbance condition, which is consistent with the result of the previous overall robustness and class-level sensitivity.

4.5 Representation Shift and Performance Degradation



(a) MFCC (b) YAMNet

Figure 8: Relationship between mean clean-to-perturbed representation shift and Macro-F1 degradation across perturbation conditions.

Figure 8 shows the relationship between the average representation shift and Macro-F1 drop under the different disturbance conditions. For MFCC, Pearson correlation is 0.949; For YAMNet, Pearson correlation is 0.978; which indicates that the degree of change in the two representations is strongly positively correlated with the decline in classification performance.

From the scattered distribution, milder noise, reverberation, and channel conditions are usually located in the region of relatively small representation distance and Macro-F1 drop, and the severe channel distortion produces the largest representation shift and most obvious performance drop.

This trend exists in the MFCC and YAMNet.

These results show that clean-to-perturbed representation change can be used as an effective indicator of the classification-performance degradation under the setting of this experiment. The Spearman correlations are approximately 0.95 for both representations.

5 Discussion

5.1 Answer to the Main Research Question

The main objective of this project is to compare the robustness of MFCC and pre-trained YAMNet representations under different recording and acoustic disturbance conditions. The experimental result shows that the performance of YAMNet has a clearer robustness advantage under the condition of background noise and severe channel distortion, but the Macro-F1 drops of the two systems are close under the reverberation condition. Therefore, YAMNet does not have the same degree of advantage under all perturbation types, but its robust performance shows a clear perturbation-dependent pattern.

5.2 Interpretation of Robustness and Representation Shift

MFCC directly relies on the local spectral envelope, so its representation changes noticeably when noise or frequency response is altered, and its statistical features can also change substantially. However, the pre-trained embeddings of YAMNet can retain higher-level sound information to some extent, so it is more stable under the condition of noise and bandwidth restriction. At the same time, the clean-to-perturbed distances of the two representations have a strong positive correlation with the Macro-F1 drop, which indicates that relatively larger representation shift is usually accompanied by more significant performance degradation.

5.3 Class-Level Findings and Comparison with Previous Work

The Class-level analysis shows that the sensitivity of different sound categories to disturbances is not consistent, and the main failure classes of the MFCC-SVM and YAMNet-SVM are not exactly the same. This demonstrates that robustness depends not only on the representation type, but also on the spectral and temporal structure of each sound category. This result is consistent with the existing research trend that pretrained audio embeddings exhibit different stabilities under channel effects and acoustic perturbations, and further demonstrates that the representation distance and downstream performance can be the complementary robustness indicators.

5.4 Limitations and Future Work

This project still has some limitations. The experiment only uses the UrbanSound8K dataset, a single pre-trained YAMNet representation and three kinds of controlled disturbance, so the result cannot directly extend to all datasets and real recording environments. Future work can include more pretrained audio models, real room impulse responses and different kinds of devices or microphone responses in the future work, and further study whether the representation shift can be used to detect in advance the potential failure of the model under unknown acoustic conditions.

6 Conclusion

This project compares the robustness of the traditional MFCC statistics and pre-trained YAMNet embeddings under different acoustic and recording conditions. The results show that the stability of the two representations is closely related to the specific perturbation type.

Under the background noise condition, the performance degradation of the YAMNet-SVM is much smaller than the MFCC-SVM; under the reverberation condition, the absolute Macro-F1 drops of

the two systems are relatively close; but the MFCC-SVM has the most obvious degradation under the severe channel distortion, and YAMNet-SVM still keeps higher classification performance.

The Class-level analysis further shows that different sound categories show clear differences in sensitivity to disturbance, and the main failure classes of the MFCC-SVM and YAMNet-SVM are not exactly the same. At the same time, the clean-to-perturbed representation shift and Macro-F1 degradation have a relatively strong positive correlation in the two representations, which indicates that the representation change can be used as a useful indicator for the model performance degradation.

In conclusion, the pre-trained YAMNet embeddings are more stable than the MFCC statistics under the background noise and severe frequency-response distortion, but this advantage is not the same for all perturbations. For real-world environmental sound classification systems, the selection of representation does not rely only on the clean-condition accuracy, but should also take into account its robustness and representation stability under the expected changes in recording conditions simultaneously.

AI Declaration

Generative artificial intelligence tools were used to assist in the language polishing, grammar modification, content organization of this report, as well as the writing, modification and debugging of some MATLAB codes. AI tools are also used to assist in checking some experimental processes and result expressions.

The experimental protocol, data processing flow, model training, performance evaluation, chart generation, and final result analysis were all run, checked, and verified by the author. The experimental data and numerical results used in the report are all derived from the actual code operation results and have not been generated by AI or fabricated experimental results. The final report content is reviewed and the author is responsible for it.

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